

Exercise 5: Task Management System

A Linked List is a linear data structure where elements are connected using links instead of being stored in contiguous memory locations.

Singly Linked List contains data and a reference to the next node.

Doubly Linked List contains data, a reference to the next node, and a reference to the previous node.

Linked Lists are useful when the size of the data changes frequently because memory is allocated dynamically.

2. Implementation

```
class Task{
    int taskID;
    String taskName;
    String status;
    public Task(int taskID, String taskName, String status){
        this.taskID = taskID;
        this.taskName = taskName;
        this.status = status;
    }
}
class Node{
    Task task;
    Node next;
    public Node(Task task){
        this.task = task;
        next = null;
    }
}
class TaskManagementSystem{
    Node head;
    public void addTask(Task task){
        Node newNode = new Node(task);
        if(head == null){
            head = newNode;
            return;
        }
        Node temp = head;
        while(temp.next != null){
            temp = temp.next;
        }
        temp.next = newNode;
    }
}
```

```

public Task searchTask(int id){
    Node temp = head;
    while(temp != null){
        if(temp.task.taskID == id){
            return temp.task;
        }
        temp = temp.next;
    }
    return null;
}

public void traverseTasks(){
    Node temp = head;
    while(temp != null){
        System.out.println(temp.task.taskID + " " + temp.task.taskName);
        temp = temp.next;
    }
}

public void deleteTask(int id){
    if(head == null){
        return;
    }
    if(head.task.taskID == id){
        head = head.next;
        return;
    }
    Node temp = head;
    while(temp.next != null){
        if(temp.next.task.taskID == id){
            temp.next = temp.next.next;
            return;
        }
        temp = temp.next;
    }
}
}

```

Analysis:

Add Task:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Search Task:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Traverse Tasks:

- Best Case: $O(n)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Delete Task:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Advantages of Linked Lists: Linked Lists can grow or shrink dynamically during program execution. Unlike arrays, they do not require a fixed size and insertion or deletion operations can be performed without shifting elements. Therefore, Linked Lists are more suitable when the amount of data changes frequently.